Bayesian Logistic Regression

using "Overeducation in the Labor Market" data
(Sicherman, *J.Lab.Econ.*, 1991*)*

Kijin Kim

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Bayesian Logit Model

- Logit Model

$$y_i \sim Bernoulli(\pi_i)$$

$$\text{logit}(\pi_i) = \log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \sum_{j=1}^{p} \beta_j x_i$$

$$f(\mathbf{y}|\boldsymbol{\beta}) = \prod_{i=1}^{n} \left(\frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{y_i} \left(\frac{1}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{1-y_i} \quad \beta_j \sim N(\mu_{\beta_j}, \sigma^2_{\beta_j})$$

$$f(\boldsymbol{\beta}|\mathbf{y}) = \frac{f(\mathbf{y}|\boldsymbol{\beta})f(\boldsymbol{\beta})}{\int f(\mathbf{y}|\boldsymbol{\beta})f(\boldsymbol{\beta})d\boldsymbol{\beta}}$$

$$= \frac{\prod_{i=1}^{n} \left(\frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{y_i} \left(\frac{1}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{1-y_i} e^{-\frac{1}{2}\sum_{j=0}^{p}\left(\frac{\beta_j-\mu_{\beta_j}}{\sigma_{\beta_j}}\right)^2}}{\int \left(\prod_{i=1}^{n} \left(\frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{y_i} \left(\frac{1}{1+e^{\mathbf{x}_i'\boldsymbol{\beta}}}\right)^{1-y_i} e^{-\frac{1}{2}\sum_{j=0}^{p}\left(\frac{\beta_j-\mu_{\beta_j}}{\sigma_{\beta_j}}\right)^2}\right) d\boldsymbol{\beta}}$$

- WHY bother to use the Bayesian approach?
- MLEs are biased for small samples (Anderson & Richardson (1979), Griffiths et al.(1987)).

- MCMC: Gibbs Sampling

1. Set initial values $\boldsymbol{\beta}^{(\mathbf{0})}$
2. For $t = 1, \cdots, T$, repeat the following steps:
   (a) $\boldsymbol{\beta} = \boldsymbol{\beta}^{(\boldsymbol{t-1})}$
   (b) For $j = 0, 1, \cdots, p$, update $\beta_j \sim f(\beta_j|\beta_{-j}, \mathbf{y})$ where $\beta_{-j} = (\beta_0, \cdots, \beta_{j-1}, \beta_{j+1}, \cdots, \beta_p)$
   (c) Set $\boldsymbol{\beta}^{(\boldsymbol{t})} = \boldsymbol{\beta}$ and save it as the generated set of values at $t+1$ iteration

Given a particular state t of the chain, we generate the new parameter values by

$$\beta_0^{(t)} \text{ from } f(\beta_0|\beta_1^{(t-1)}, \cdots, \beta_p^{(t-1)}, \mathbf{y})$$
$$\beta_1^{(t)} \text{ from } f(\beta_1|\beta_0^{(t)}, \beta_2^{(t-1)}, \cdots, \beta_p^{(t-1)}, \mathbf{y})$$
$$\beta_2^{(t)} \text{ from } f(\beta_2|\beta_0^{(t)}, \beta_1^{(t)}, \beta_3^{(t-1)}, \cdots, \beta_p^{(t-1)}, \mathbf{y})$$
$$\vdots$$
$$\beta_p^{(t)} \text{ from } f(\beta_p|\beta_0^{(t)}, \cdots, \beta_{p-1}^{(t)}, \mathbf{y})$$

MCMC: Metropolis-Hastings

- Since $f(\beta_j \mid \beta_{-j}, y)$ is not a simple distribution, a standard random number generator might not work.
- The MH is an alternative, but multi-variate random-walk approach is highly inefficient when the parameters are correlated.
- Gamerman (1997) proposed an algorithm combining the MH and the IWLS proposal density (the default in WinBUGS)

- MH algorithm

1. Set $\boldsymbol{\beta}^{(0)}$
2. For $t=1,\cdots,T$, repeat the following steps
   (a) Set $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t-1)}$
   (b) Generate $\tilde{\boldsymbol{\beta}}$ from a proposal density $q(\tilde{\boldsymbol{\beta}}|\boldsymbol{\beta})$
   (c) Calculate $\alpha = \min\left(1, \frac{f(\tilde{\boldsymbol{\beta}}|\mathbf{y})q(\tilde{\boldsymbol{\beta}}|\boldsymbol{\beta})}{f(\boldsymbol{\beta}|\mathbf{y})q(\boldsymbol{\beta}|\tilde{\boldsymbol{\beta}})}\right)$
   (d) Update $\boldsymbol{\beta}^{(t)} = \tilde{\boldsymbol{\beta}}$ with probability α

Empirical Estimation

- Data: a sample of male head of households aged 18-60 from the 1976-1979 waves of the PSID
- Research question in Sicherman (1991): "Are overeducated workers more likely to change firms and occupations?"
- My question in Sicherman's data: "What factors influence job satisfaction?"

- Firm (Occupation) Change

$$y_i \sim Bernoulli(\pi_i)$$

$$y_i = 1 \text{ if firm (or occupation) change happens; } 0 \text{ otherwise}$$

$$\begin{aligned}\text{logit}(\pi_i) = \beta_0 &+ \beta_1 \text{Schooling}_i + \beta_2 \text{Exp}_i + \beta_3 \text{Exp}_i^2 + \beta_4 \text{Union}_i \\ &+ \beta_5 \text{Black}_i + \beta_6 \text{LargeCity}_i + \beta_7 \text{Married}_i + \beta_8 \text{Disabled}_i \\ &+ \beta_9 \text{Overschooling}_i + \beta_{10} \text{Underschooling}_i\end{aligned}$$

- Job Satisfaction

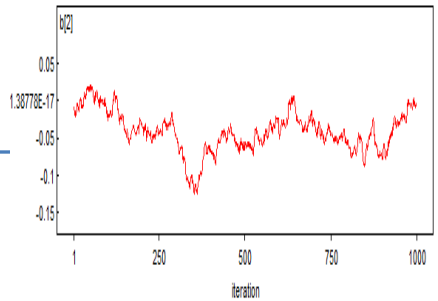
$$y_i \sim Bernoulli(\pi_i)$$

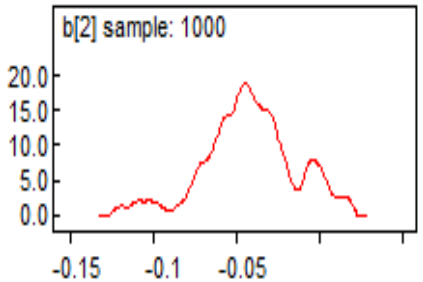
$$y_i = 1 \text{ if the current job is considered to be better; } 0 \text{ otherwise}$$

$$\begin{aligned}\text{logit}(\pi_i) = \beta_0 &+ \beta_1 \text{Overschooling} + \beta_2 \text{Underschooling}_i \\ &+ \beta_3 \text{Wage Diff}_i + \beta_4 \text{Occupation Diff}_i \\ &+ \beta_5 \text{Schooling}_i + \beta_6 \text{Exp}_i + \beta_7 \text{Exp}_i^2 + \beta_8 \text{Union}_i \\ &+ \beta_{12} \text{RequiredTraining}_i\end{aligned}$$

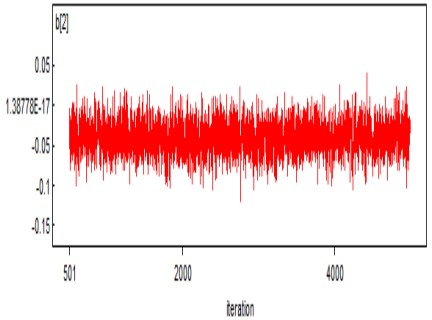
MLE vs. MCMC – Large Sample

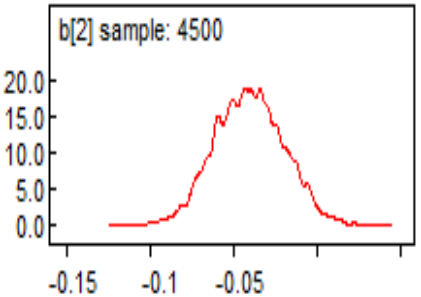
1,000 iterations





5,000 iterations, 500 burn-in, 50 thinning



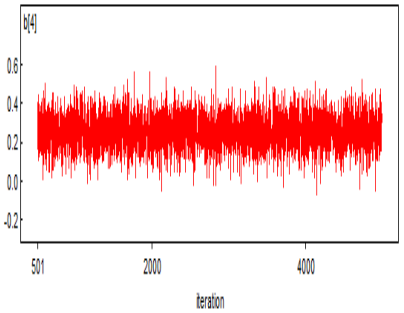


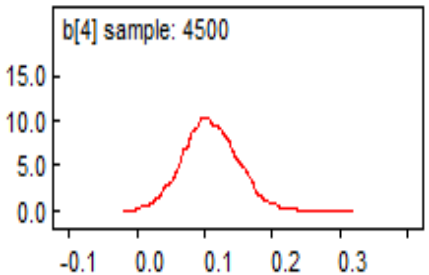
	Firm Change (N=2,441)				Occupation Change (N=3,484)			
	MLE		MCMC		MLE		MCMC	
	Coef.[1)]		Mean[2)]		Coef.[1)]		Mean[2)]	
Intercept	-0.5406	(1.5)	-0.5505	(1.5)	0.6999	(3.1)	0.7026	(3.1)
Schooling	-0.0697	(3.2)	-0.0698	(3.2)	-0.0619	(4.5)	-0.0621	(4.6)
Experience	-0.0404	(1.9)	-0.0403	(1.9)	-0.0338	(2.8)	-0.0335	(2.7)
Experience^2	0.0006	(1.1)	0.0006	(1.0)	0.0001	(0.5)	0.0001	(0.4)
Union	-0.4057	(3.0)	-0.4105	(3.1)	0.0544	(0.7)	0.0562	(0.7)
Race (1=black)	0.3103	(2.4)	0.3123	(2.4)	0.1326	(1.6)	0.1322	(1.6)
Large City	-0.1906	(1.5)	-0.1935	(1.5)	-0.0569	(0.7)	-0.0583	(0.7)
Married	-0.1486	(0.9)	-0.1452	(0.9)	0.0597	(0.6)	0.0585	(0.6)
Disabled	0.2988	(1.3)	0.2885	(1.3)	0.0505	(0.4)	0.0504	(0.4)
Overschooling	0.4418	(3.3)	0.4426	(3.3)	0.1845	(2.4)	0.1842	(2.4)
Underschooling	0.0249	(0.1)	0.0190	(0.1)	0.2570	(2.2)	0.2561	(2.2)

Note: 1) Asymptotic (absolute) t-statistics are in parentheses.; 2) Absolute values of Mean/S.d are in parentheses.
* Overschooling = 1 if years of schooling > required schooling (0 otherwise); Underschooling = 1 if years of schooling < required schooling (0 otherwise)

MLE vs. MCMC – Small Sample

5,000 iterations, 500 burn-in, 50 thinning





- Small sample bias in MLEs
- The MLE could possibly lead to a wrong conclusion.

	MLE		MCMC (N=181)				
	Coef.[1)]		Mean[2)]		2.5%Q	Median	97.5%Q
Intercept	2.8102	(1.75)	3.1510	(1.86)	-0.2320	3.1490	6.4290
Overschooling	-0.6141	(1.19)	-0.7099	(1.31)	-1.7930	-0.6927	0.3198
Underschooling	1.1949	(0.94)	1.7370	(1.12)	-0.7672	1.5730	5.2930
Wage Diff.	0.1522	(1.90)	0.1702	(1.98)	0.0080	0.1690	0.3434
Occupation Diff.	0.0962	(2.62)	0.1089	(2.76)	0.0334	0.1075	0.1883
Schooling	0.0523	(0.53)	0.0611	(0.58)	-0.1430	0.0606	0.2715
Experience	0.0412	(0.39)	0.0438	(0.39)	-0.1785	0.0447	0.2623
Experience2	-0.0025	(0.75)	-0.0027	(0.73)	-0.0100	-0.0027	0.0046
Union	0.0117	(0.02)	0.0641	(0.10)	-1.1580	0.0516	1.3770
Race (1=black)	-0.4135	(0.83)	-0.4312	(0.82)	-1.4280	-0.4346	0.5977
Large City	-0.4829	(0.93)	-0.5176	(0.94)	-1.6020	-0.5258	0.5676
Married	-0.9582	(1.33)	-1.1570	(1.48)	-2.8090	-1.1140	0.2951
Disabled	-0.3092	(0.34)	-0.2064	(0.21)	-2.0720	-0.2697	1.9130
Required Training	-0.1120	(1.01)	-0.1140	(0.94)	-0.3462	-0.1159	0.1327

Note: 1) Asymptotic (absolute) t-statistics are in parentheses.; 2) Absolute values of Mean/S.d are in parentheses.; 3) Dependent variable = whether satisfied with the present job

Empirical 95% CI=[0.008,0.343]

p-value = 0.058